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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	404notFound
Selected Domain	Domain 2- Orbyte Smart SME Challenge
Selected SDG Track (1–17)	SDG 12 – Responsible Consumption and Production
Project / Solution Name	StockSenseAI
Name (Leader)	Haemaavadhaana Seemadaran
Email & Phone (Leader)	Email: haemaavadhaana@gmail.com Phone: 0182854952
University Affiliation(s)	Asia Pacific University of Technology and Innovation
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium-sized enterprises (SMEs), particularly independent mini-markets and grocery shops in Malaysia, face challenges in managing inventory efficiently. Many small retailers rely on manual stock counting, spreadsheets, or basic inventory systems to monitor product availability and determine when to replenish stock. These approaches can make it difficult to identify low-stock items promptly, maintain accurate inventory records, and make informed purchasing decisions.

Inefficient inventory management can lead to two major problems: stockouts and overstocking. Stockouts result in lost sales, dissatisfied customers, and reduced customer loyalty when essential products are unavailable. Overstocking ties up limited working capital, occupies valuable storage space, and increases the risk of products expiring or becoming unsellable, particularly for perishable goods. These challenges place additional pressure on SME owners and employees, who often have limited time and resources to manage daily operations.

The problem also has environmental and economic implications. Excess inventory, especially perishable products, can contribute to avoidable waste, while unnecessary replenishment and inefficient stock handling consume additional resources. Without accessible tools to support timely decisions, smaller retailers may struggle to achieve the inventory efficiency of larger businesses with more advanced management systems.

Therefore, there is a need for an affordable, user-friendly solution that helps SMEs monitor inventory, identify replenishment needs, and manage purchasing decisions more systematically. Improving inventory visibility and replenishment workflows can support better resource utilisation, reduce avoidable waste, and strengthen the operational resilience of small retailers.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

StockSenseAI is an AI-assisted inventory management solution designed to help Malaysian mini-markets and small retailers improve stock visibility and replenishment decisions. The application aims to simplify inventory monitoring, identify products requiring replenishment, and organise purchasing decisions through a structured approval workflow.

The current prototype includes inventory management, replenishment requests, an Approval Centre, and an activity log. Users can manage product stock, create replenishment requests for items such as printer ink, review pending requests, approve or reject them, and track recorded actions. These features provide a more organised approach to inventory control and purchasing oversight.

StockSenseAI is designed to support data-informed decision-making by using inventory information to highlight low-stock items and guide replenishment recommendations. AI capabilities can be developed further to analyse historical stock movements, estimate future demand, and recommend appropriate reorder quantities. These capabilities would help retailers make more informed decisions and reduce the risks of stockouts and excess inventory. Advanced forecasting will be treated as a planned enhancement unless it has been implemented and tested.

The project uses Python for application logic and inventory workflows, with a web-based development and hosting environment through Replit. The system is organised into reusable components to support maintainability and future integration. Database architecture will support the structured storage of inventory records, replenishment requests, and approval activity.

Building on the SDC Phase 1 workshops, the team intends to apply relevant skills in generative AI APIs, database design, and cloud deployment where appropriate. These technologies will support the development of a functional, accessible prototype that can be tested and improved for small-business use. Any sponsor-provided APIs or cloud platforms will be identified according to actual usage.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

StockSenseAI is developed using Python, with Replit serving as the development environment and intended deployment platform. The application uses modular components to manage inventory records, replenishment requests, approval workflows, and activity logging.

The conceptual technology stack consists of a Python-based application layer for inventory processing and business logic, a user interface for interacting with inventory and approval functions, and a data storage layer for maintaining product information, replenishment requests, and activity records. Inventory logic identifies items requiring attention, while the approval workflow allows users to review, approve, or reject replenishment requests. The activity log records workflow actions to support traceability.

The proposed architecture is:

User Interface → Python Application Logic → Inventory and Replenishment Workflows → Data Storage → Results displayed to the user.

Future enhancements may incorporate a generative AI API to provide natural-language inventory insights and recommendations, alongside a structured database and cloud deployment configuration. These additions will be implemented and tested as appropriate for the Phase 2 prototype.

The project will use relevant skills from the SDC workshops, including application development, database architecture, API integration, and cloud deployment. No official SDC sponsor-provided GenAI API or cloud platform is claimed as utilised unless its use has been confirmed.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

StockSenseAI follows a user-focused development approach by addressing common inventory management challenges faced by small retailers. The solution prioritises practical workflows that help users monitor stock, submit replenishment requests, review purchasing decisions, and maintain an activity record.

The prototype has been developed incrementally, with reusable components for inventory management and workflow processing. Functional verification includes creating a replenishment request for an inventory item, confirming that the request appears in the Approval Centre, approving or rejecting the request, and checking that the activity log records the action. These checks help validate the core workflow before further development.

For Phase 2, the team intends to improve usability through feedback and iterative testing, strengthen data management, and prepare the application for reliable deployment. Planned work includes validating inventory calculations, handling invalid inputs, testing approval workflows, and reviewing accessibility and ease of use. Database configuration, security controls, and deployment requirements will also be assessed before broader testing with representative SME users.

This iterative approach aims to produce a practical, maintainable, and user-friendly application that can be evaluated against measurable operational and sustainability outcomes.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	UN SDG 12, Target 12.3: By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Baseline to be established through testing with a representative inventory dataset before evaluating the solution. Initial measurements will record the number of stockout events, excess-stock items, and expired or wasted products over a defined observation period. The baseline will be established before comparing results with StockSenseAI-assisted inventory management.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in stockout events, measured by comparing the number of stockout events before and after applying StockSenseAI-assisted replenishment decisions to the same test inventory over equivalent observation periods.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target a 15% reduction in stockout events compared with the established baseline during Phase 2 testing, measured using the same inventory dataset and equivalent test conditions. This is a proposed target to be evaluated, not a verified result.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [√] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [√] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [√] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [√] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ◉ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

◉ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type(code/dataset/model)**	**License**
Python	Programming language/runtime	Python Software Foundation License
Streamlit	Open-source framework	Apache License 2.0
Pandas	Code library	BSD 3-Clause License
Pandas *(if used)*	Code library	BSD 3-Clause License

6.3 Employment-Related IP Conflicts

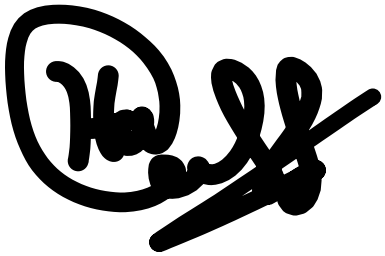
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ◉ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

23rd October 2026